Enthalpy-Primary Stiff Phase Change, Adaptive Time Stepping, and Discrete Adjoints Beyond Pennes Perfusion in Thermalio

Thermalio research notes — follow-up to Directions C and D

Abstract

This report documents three capabilities added to Thermalio's polygonal finite-volume (FV) heat-transfer framework and the evidence that they are correct. (i) An *enthalpy-primary* formulation of the (optionally non-Fourier, Cattaneo) Stefan problem, solved by a globalized Newton method in the enthalpy variable; it is discretely energy conservative to round-off and its Jacobian is uniformly bounded in the latent heat and mushy-zone width. (ii) *Adaptive time stepping* with an extrapolation-based local-error estimator, an elementary step-size controller, failure-driven step reduction and exact output-time landing. (iii) *Exact discrete adjoints* for conductivity, perfusion and source *fields*, Robin heat-transfer coefficients, the Caputo L1 scheme (fractional order β), and the nonlinear enthalpy Stefan / Cattaneo–Stefan solver (latent heat, diffusivity, relaxation time) on variable time steps. Manufactured solutions confirm first-order temporal and second-order spatial convergence; a 35-case stiffness sweep shows that the apparent-heat-capacity Picard scheme fails in 20 cases and violates energy conservation by up to 0.67 energy units, while the enthalpy scheme never fails and conserves energy to 1.5×10^{-10}; on a laser-pulse/cooling problem adaptive stepping is 10–120× more accurate than uniform stepping at comparable Newton work. All six adjoints pass Taylor tests with second-order remainders; a 50-parameter conductivity field and a latent-heat/diffusivity pair are recovered from sparse noisy sensors.

Contents

1 Overview and summary of results **2**

2 Spatial discretization (common to all solvers) **2**

3 Enthalpy-primary formulation of stiff phase change **3**
3.1 Continuous model 3
3.2 Inverting the enthalpy law 4
3.3 Variable-step time discretization 4
3.4 Newton's method and why the enthalpy variable is robust 5
3.5 Exact discrete energy conservation 5

4 Adaptive time stepping **6**
4.1 Local error estimator 6
4.2 Controller 6

5 Discrete adjoints **7**
5.1 General framework 7
5.2 Linear parabolic adjoint: conductivity, perfusion and source fields, Robin coefficient 7
5.3 Fractional (Caputo L1) adjoint 8
5.4 Nonlinear enthalpy (Stefan / Cattaneo–Stefan) adjoint 8

6 Numerical results **9**
6.1 Manufactured-solution verification of the enthalpy solver 9
6.2 Robustness for stiff mushy zones and energy conservation 9
6.3 Adaptive time stepping on a multi-scale pulse problem 10
6.4 Adjoint verification and cost 10
6.5 Conductivity-field inversion 11
6.6 Nonlinear parameter recovery 11

7 Limitations and next steps **12**

8 Reproducibility **13**

1 Overview and summary of results

Direction C of the Thermalio research backlog introduced apparent-heat-capacity models of non-Fourier and fractional phase change and identified, among its open items, *adaptive time stepping for stiff phase change* and an *enthalpy-primary nonlinear formulation.* Direction D introduced an inverse-problem testbed whose only analytic gradient was a discrete adjoint for a scalar Pennes perfusion rate; it listed *adjoints for diffusivity, source amplitudes, Robin boundary parameters, Cattaneo/fractional transport and nonlinear phase change* as the next step. This report covers both follow-ups.

Table 1: Implemented components. All live in the `heat_solver` package.

Component	Location
Smooth enthalpy law $H = cT + L\phi(T)$, safeguarded inverse	`enthalpy.SmoothPhaseChangeModel`
Enthalpy-primary (Cattaneo–)Stefan solver, Newton	`enthalpy.EnthalpyStefanSolver`
Adaptive step control	`enthalpy.AdaptiveStepOptions`
Apparent-capacity Picard reference solver	`enthalpy.ApparentCapacityPicardSolver`
Field / Robin / source adjoint (backward Euler)	`adjoint.LinearHeatAdjoint`
Caputo L1 adjoint (β, α)	`adjoint.fractional_heat_adjoint`
Nonlinear enthalpy adjoint (L, α, τ)	`adjoint.enthalpy_stefan_adjoint`
Taylor-remainder gradient test	`adjoint.taylor_test`
Studies / figures	`direction_c_enthalpy_studies.py` `direction_d_adjoint_studies.py`
Unit tests	`tests/test_enthalpy.py` `tests/test_adjoint.py`

2 Spatial discretization (common to all solvers)

Let $\Omega \subset \mathbb{R}^2$ be partitioned into polygonal cells Ω_i, $i = 1, \dots, M$, with areas $|\Omega_i|$ and centroids $\boldsymbol{x}_i$. Write $\mathsf{D} = \mathrm{diag}(|\Omega_i|)$. For an isotropic conductivity $\kappa(\boldsymbol{x})$ the two-point flux approximation (with the optional non-orthogonal gradient correction used throughout Thermalio) gives a diffusion matrix that is *linear in the face conductivities*:

$$\mathsf{A}(\kappa) = \sum_{e \in \mathcal{E}_{\mathrm{int}}} \kappa(\boldsymbol{x}_e)\, \mathsf{G}_e, \qquad (\mathsf{G}_e)_{ij} \text{ geometric only}, \tag{2.1}$$

where $\boldsymbol{x}_e$ is the midpoint of interior face e. On an orthogonal face $\mathsf{G}_e = \frac{|e|}{d_e}(\boldsymbol{e}_i - \boldsymbol{e}_j)(\boldsymbol{e}_i - \boldsymbol{e}_j)^\top$, with d_e the normal distance between centroids, so A is symmetric with zero row sums and

non-positive off-diagonals (an M-matrix up to its kernel). Equation (2.1) is the key structural fact used by the field adjoint (section 5.2).

Boundary conditions follow Thermalio's conventions.

- *Dirichlet*: the equations of boundary cells $\mathcal{B}$ (cells touching $\partial\Omega$) are replaced by $u_i = g(\boldsymbol{x}_i, t)$; the interior index set is $\mathcal{I} = \{1..M\} \setminus \mathcal{B}$.
- *Prescribed inward flux*: $b_i = \sum_{f \subset \partial\Omega_i \cap \partial\Omega} |f|\, q_{\text{in}}(\boldsymbol{x}_f, t)$ enters the right-hand side; $\mathcal{I} = \{1..M\}$.
- *Robin* $\kappa\partial_n u + h(u - u_\infty) = 0$: eliminating the face value with the one-sided distance d_f gives the face coefficient

$$\epsilon_f(\kappa_f, h) = \frac{\kappa_f h}{\kappa_f + h d_f}, \qquad \mathsf{B} = \operatorname{diag}\Big(\textstyle\sum_f |f|\,\epsilon_f\Big), \quad b_i = \textstyle\sum_f |f|\,\epsilon_f u_\infty. \tag{2.2}$$

We write $\mathsf{K} = \mathsf{A} + \mathsf{B}$ for the full linear operator.

3 Enthalpy-primary formulation of stiff phase change

3.1 Continuous model

The enthalpy method for phase change goes back to fixed-grid formulations such as [1, 2], with convergence theory for the classical Stefan problem in [3]; here it is combined with the Cattaneo flux law [4, 5] and with Thermalio's polygonal FV operators.

Energy conservation and the Maxwell–Cattaneo–Vernotte flux law read

$$\partial_t H + \nabla \cdot \boldsymbol{q} = Q, \qquad \tau\, \partial_t \boldsymbol{q} + \boldsymbol{q} = -\kappa \nabla T, \tag{3.1}$$

with volumetric enthalpy H, temperature T and relaxation time $\tau \geq 0$. For time-independent κ and τ, applying $(1 + \tau\partial_t)$ to the first equation and eliminating $\boldsymbol{q}$ yields the *hyperbolic enthalpy equation*

$$\tau\, \partial_{tt} H + \partial_t H - \nabla \cdot (\kappa \nabla T) = \tilde{Q}, \qquad \tilde{Q} = Q + \tau \partial_t Q, \qquad T = \mathcal{T}(H). \tag{3.2}$$

($\tilde{Q}$ is what the solver receives as its source; for $\tau = 0$ it is just Q.) The constitutive enthalpy law is

$$H(T) = cT + L\,\phi(T), \qquad c_{\text{app}}(T) = H'(T) = c + L\phi'(T), \tag{3.3}$$

with specific heat $c > 0$, latent heat $L \geq 0$ and liquid fraction $\phi : \mathbb{R} \to [0, 1]$ nondecreasing. Two laws are supported:

$$\text{ramp:} \quad \phi(T) = \operatorname{clip}\big((T - T_s)/(T_\ell - T_s), 0, 1\big), \tag{3.4}$$

$$\text{smooth:} \quad \phi(T) = \tfrac{1}{2}\big(1 + \tanh((T - T_m)/\delta)\big), \qquad \phi' = \tfrac{1}{2\delta}\operatorname{sech}^2\big(\tfrac{T - T_m}{\delta}\big). \tag{3.5}$$

The capacity spike has height $c_{\text{app}}^{\max} = c + L/(2\delta)$ (smooth) or $c + L/(T_\ell - T_s)$ (ramp); the *stiffness* of the problem is governed by L/δ.

Remark 3.1 (Relation to the apparent-capacity Cattaneo–Stefan model). Thermalio's existing `HyperbolicStefanSolver` discretizes $\tau T_{tt} + c_{\text{app}}(T)T_t - \nabla\cdot(\kappa\nabla T) = Q$. Since $H_{tt} = c_{\text{app}}(T)T_{tt} + c'_{\text{app}}(T)T_t^2$, the enthalpy model (3.2) differs by $\tau[(c_{\text{app}} - 1)T_{tt} + c'_{\text{app}}T_t^2]$. Both coincide for $L = 0$, $c = 1$ (verified to 10^{-15} in the unit tests); the enthalpy form is the one that follows from conservation plus the Cattaneo law and is therefore the physically consistent extension.

3.2 Inverting the enthalpy law

Because H is strictly increasing, $\mathcal{T} = H^{-1}$ exists and is Lipschitz:

$$\mathcal{T}'(H) = \frac{1}{c_{\mathrm{app}}(\mathcal{T}(H))} \in \Big(0, \frac{1}{c}\Big]. \tag{3.6}$$

For the ramp law $\mathcal{T}$ is piecewise linear and inverted in closed form. For the smooth law, $0 \le \phi \le 1$ gives the bracket $\mathcal{T}(H) \in [(H-L)/c,\ H/c]$; we start from the inverse of the equivalent ramp on $[T_m - \delta, T_m + \delta]$ and run a vectorized safeguarded Newton iteration that bisects whenever the Newton iterate leaves the current bracket or fails to halve $|H(T) - H|$ ("`rtsafe`"). Plain Newton is *not* sufficient: on the latent plateau it bounces between the two flat branches and converges only linearly (we observed $O(1)$ inversion errors before adding the halving test). With the safeguard the inversion error is below 3×10^{-13} for $\delta \in [10^{-4}, 0.5]$, $L \le 50$.

3.3 Variable-step time discretization

Let $t_{n+1} = t_n + k_n$. We use the variable-step three-level operators

$$D_n^2 H = \frac{2}{k_n + k_{n-1}}\Big[\frac{H^{n+1} - H^n}{k_n} - \frac{H^n - H^{n-1}}{k_{n-1}}\Big], \tag{3.7}$$

$$D_n^1 H = \begin{cases} (H^{n+1} - H^n)/k_n, & \tau = 0 \text{ (backward Euler)}, \\ (H^{n+1} - H^{n-1})/(k_n + k_{n-1}), & \tau > 0 \text{ (centered)}, \end{cases} \tag{3.8}$$

and seek H^{n+1} such that, on interior rows $i \in \mathcal{I}$,

$$R^{n+1}(H^{n+1}) := \mathsf{D}\big[\tau D_n^2 H + D_n^1 H - \tilde{Q}^{n+1}\big] + \mathsf{K}\,\mathcal{T}(H^{n+1}) - b^{n+1} = 0, \tag{3.9}$$

and $R_i^{n+1} = H_i^{n+1} - H(g_i^{n+1})$ on Dirichlet rows. Collecting coefficients,

$$\tau D_n^2 H + D_n^1 H = \gamma_n H^{n+1} - p_n H^n - s_n H^{n-1}, \tag{3.10}$$

$$\gamma_n = \frac{2\tau}{S_n k_n} + \begin{cases} \frac{1}{k_n} \\ \frac{1}{S_n} \end{cases}, \quad p_n = \frac{2\tau}{S_n}\Big(\frac{1}{k_n} + \frac{1}{k_{n-1}}\Big) + \begin{cases} \frac{1}{k_n} \\ 0 \end{cases}, \quad s_n = -\frac{2\tau}{S_n k_{n-1}} + \begin{cases} 0 \\ \frac{1}{S_n} \end{cases}, \tag{3.11}$$

with $S_n = k_n + k_{n-1}$ and upper/lower cases for $\tau = 0$ / $\tau > 0$. For $\tau > 0$ the first step needs H^{-1}; with $v^0 = c_{\mathrm{app}}(T^0)\,\partial_t T^0$ and $a^0 = \big(\tilde{Q}^0 + \mathsf{D}^{-1}(b^0 - \mathsf{K}T^0) - v^0\big)/\tau$ (the PDE at t_0),

$$H^{-1} = H^0 - k_0 v^0 + \tfrac{1}{2} k_0^2 a^0, \qquad k_{-1} := k_0. \tag{3.12}$$

On uniform steps and $L = 0$ the scheme is *identical* to the existing `ReactionDiffusionHeatSolver` (backward Euler) and `HyperbolicHeatSolver` (three-level), which we verify to machine precision.

Remark 3.2 (Temporal order of the three-level Cattaneo scheme). D_n^2 and the centered D_n^1 are second-order accurate at t_n, but $\mathsf{K}T$ and $\tilde{Q}$ are evaluated at t_{n+1}, so the truncation error is $k(\mathsf{K}\,\partial_t T - \partial_t \tilde{Q}) + O(k^2)$: the scheme is *first order* in time. We measured order 1.01–1.03 for the pre-existing `HyperbolicHeatSolver` (whose documentation called it second order; its regression test refines h and k together and only checks error reduction). The documentation has been corrected. A second-order variant would evaluate K at the weighted average $\frac{1}{4}(T^{n+1} + 2T^n + T^{n-1})$ (Newmark $\beta = \frac{1}{4}$); this is left as future work.

3.4 Newton's method and why the enthalpy variable is robust

The Jacobian of (3.9) is

$$\mathsf{J}_n(H) = \gamma_n \mathsf{D} + \mathsf{K}\,\mathsf{S}(H), \qquad \mathsf{S}(H) = \operatorname{diag}\big(\mathcal{T}'(H_i)\big) = \operatorname{diag}\big(1/c_{\mathrm{app}}(T_i)\big), \tag{3.13}$$

with identity rows on $\mathcal{B}$. Newton updates $\mathsf{J}_n\delta = -R$ are globalized by backtracking on $\|R\|_2$ (Armijo constant 10^{-4}); iterations stop when $\|R\|_\infty \le 10^{-10}\,\|\gamma_n \mathsf{D} H\|_\infty$ or $\|\delta\|_\infty \le 10^{-10}\max(1, \|H\|_\infty)$.

Proposition 3.3 (Well-posed, L-uniformly bounded Newton systems)**.** *Assume* K *is a Z-matrix with non-negative row sums (true for two-point fluxes on* K*-orthogonal meshes with Dirichlet, flux or Robin closures). Then for every* H *the Jacobian* (3.13) *is nonsingular and*

$$\|\mathsf{J}_n(H) - \gamma_n \mathsf{D}\|_\infty \le \frac{\|\mathsf{K}\|_\infty}{c}, \tag{3.14}$$

independently of L *and of the mushy-zone width.*

Proof. $\mathsf{J}_n = (\gamma_n \mathsf{D}\mathsf{S}^{-1} + \mathsf{K})\mathsf{S}$. The first factor is a Z-matrix whose diagonal dominates its off-diagonal row sums strictly (because $\gamma_n|\Omega_i|c_{\mathrm{app}} > 0$ is added to a weakly diagonally dominant row), hence a nonsingular M-matrix; S is a positive diagonal matrix. The bound follows from $0 < \mathcal{T}' \le 1/c$ in (3.6). □

By contrast, the temperature-primary (apparent-capacity) update $\mathsf{D}c_{\mathrm{app}}(T^k)(T^{k+1}-T^n)/k_n + \mathsf{K}T^{k+1} = \dots$ has a matrix diagonal of size $|\Omega_i|c_{\mathrm{app}}/k_n = O(L/\delta)$ that *switches on and off* as cells enter and leave the mushy interval, and the nonlinearity $c_{\mathrm{app}}(T)$ has Lipschitz constant $O(L/\delta^2)$. In H the nonlinearity $\mathcal{T}$ has Lipschitz constant $1/c$ and becomes *weaker* (flatter) in the mushy zone.

Lemma 3.4 (Non-existence of an apparent-capacity fixed point for the ramp law)**.** *Consider one cell,* $\tau = 0$*, ramp law* (3.4) *with* $w = T_\ell - T_s > 0$*,* $L > 0$*, and the scalar apparent-capacity equation* $f(T) := c_{\mathrm{app}}(T)(T - T^n) = r$ *with* $T^n < T_s$*. The equation has no solution for*

$$r \in \Big(c(T_s - T^n),\ \min\big\{(c + L/w)(T_s - T^n),\ c(T_\ell - T^n)\big\}\Big],$$

a non-empty interval.

Proof. On the solid branch $T < T_s$, $f(T) = c(T - T^n) < c(T_s - T^n)$; on the mushy branch $T \in [T_s, T_\ell]$, $f(T) = (c + L/w)(T - T^n) \ge (c + L/w)(T_s - T^n)$; on the liquid branch $T > T_\ell$, $f(T) = c(T - T^n) > c(T_\ell - T^n)$. None of these sets meets the stated interval. □

The enthalpy equation $H(T) - H(T^n) = r$ has a unique solution for every r since H is continuous and strictly increasing. Lemma 3.4 is consistent with our observation that the apparent-capacity Picard iteration of the polygonal solver failed at *every* time step we tried (10^{-2} down to 5×10^{-4}) on a hot-wall melting problem with the ramp law; the comparison in section 6.2 therefore uses the smooth law, for which f is continuous and apparent-capacity solutions exist.

3.5 Exact discrete energy conservation

Proposition 3.5 (Discrete energy balance, $\tau = 0$)**.** *Let* $E^n = \sum_{i\in\mathcal{I}} |\Omega_i| H_i^n$*. Every accepted step satisfies*

$$E^{n+1} - E^n = k_n \sum_{i\in\mathcal{I}} \Big(|\Omega_i| Q_i^{n+1} + b_i^{n+1} - (\mathsf{K}T^{n+1})_i\Big) + k_n \sum_{i\in\mathcal{I}} R_i^{n+1}. \tag{3.15}$$

For flux/insulated boundaries $\sum_i (\mathsf{A}T)_i = 0$*, so the change of total enthalpy equals the injected energy* $k_n \sum_i (|\Omega_i| Q_i + b_i)$ *up to the Newton residual. For Dirichlet boundaries* $-\sum_{\mathcal{I}} (\mathsf{A}T)_i$ *is exactly the conductive flux from the boundary band into the interior.*

Proof. Sum (3.9) (with $\tau = 0$) over $i \in \mathcal{I}$ and multiply by k_n. □

The solver records the per-step balance residual (the last term) and the cumulative input; in all experiments the residual is $\le 1.5 \times 10^{-10}$ and typically 10^{-17}. This holds for *any* step sequence, so adaptivity never compromises conservation. For the apparent-capacity scheme the same bookkeeping exposes a defect

$$\begin{aligned} \mathcal{E}^n &= \sum_{i\in\mathcal{I}} |\Omega_i| \Big[H(T_i^{n+1}) - H(T_i^n) - c_{\mathrm{app}}(T_i^{n+1})(T_i^{n+1} - T_i^n) \Big] \\ &= \sum_{i\in\mathcal{I}} |\Omega_i| \int_{T_i^n}^{T_i^{n+1}} \big(c_{\mathrm{app}}(s) - c_{\mathrm{app}}(T_i^{n+1}) \big)\, \mathrm{d}s, \end{aligned} \tag{3.16}$$

which is $O(L|\Omega_i|)$ whenever a cell crosses a substantial part of the mushy interval in one step ("latent-heat jumping").

4 Adaptive time stepping

4.1 Local error estimator

For backward Euler on $H' = F(t, H)$ the local error is $e_{n+1} = H^{n+1} - H(t_{n+1}) = \frac{1}{2}k_n^2 H'' + O(k^3)$. The linear extrapolant $H^p = H^n + \frac{k_n}{k_{n-1}}(H^n - H^{n-1})$ satisfies $H(t_{n+1}) - H^p = \frac{1}{2}k_n(k_n + k_{n-1})H'' + O(k^3)$. Subtracting, $H^{n+1} - H^p = \frac{1}{2}k_n(2k_n + k_{n-1})H'' + O(k^3)$, hence

$$\widehat{e}_{n+1} = \frac{k_n}{2k_n + k_{n-1}} \big(H^{n+1} - H^p \big) = e_{n+1} + O(k^3). \tag{4.1}$$

The estimator is exact for quadratic-in-time enthalpy histories (unit test). It costs nothing beyond the step itself, and H^p also serves as the Newton initial guess.

4.2 Controller

With component-wise tolerances, the scaled error is

$$\mathrm{err}_{n+1} = \max_i \frac{|\widehat{e}_{n+1,i}|}{\mathrm{atol} + \mathrm{rtol}\, \max(|H_i^{n+1}|, |H_i^n|)}. \tag{4.2}$$

(The max norm is the default because phase fronts are localized; an RMS norm is available.) The step is accepted iff $\mathrm{err} \le 1$, and the next step is

$$k_{\mathrm{new}} = k_n \, \min\Big(f_{\max},\ \max\big(f_{\min},\ \rho\, \mathrm{err}^{-1/2} \big) \Big), \qquad \rho = 0.9,\ f_{\min} = 0.2,\ f_{\max} = 2, \tag{4.3}$$

the elementary (I-) controller for a method with local error $O(k^2)$ [7, 8]. Further rules:

1. *Failure-driven reduction*: if Newton fails, the step is retried with $k \leftarrow k/2$ (bounded by $k_{\min}$ and a maximum rejection count).
2. *Output times / breakpoints* are hit exactly (like `tstop` in CVODE). Forcing discontinuities (pulse switch-off) should be listed as output times so that a step never straddles them.
3. *Sliver avoidance*: if $k < t_{\mathrm{out}} - t < 2k$ the remaining interval is split into two equal steps. Without this rule the controller produced 3×10^{-6} steps just before output times.
4. The first step has no estimate and is accepted on Newton convergence.
5. For $\tau > 0$, (4.1) is used as a curvature indicator for the three-level scheme (which is first order on variable steps, remark 3.2), not as an asymptotically exact error estimate.

5 Discrete adjoints

5.1 General framework

Let the time-stepping scheme be $R^n(y^n, y^{n-1}, y^{n-2}; \theta) = 0$, $n = 1..N$, with given $y^0(\theta)$, $y^{-1}(\theta)$, and the misfit

$$\mathcal{J}(\theta) = \frac{1}{2} \sum_{m \in \mathcal{O}} \big(\mathsf{P}\,\eta(y^m) - d^m\big)^\top \mathsf{W}^m \big(\mathsf{P}\,\eta(y^m) - d^m\big), \tag{5.1}$$

where P selects sensor cells, W^m are weights and η maps the state to the observed quantity ($\eta(y) = y$, or $\eta(H) = \mathcal{T}(H)$ for the enthalpy solver). With the Lagrangian $\mathcal{L} = \mathcal{J} - \sum_n \lambda_n^\top R^n$, stationarity with respect to y^m gives the backward recurrence ($\lambda_{N+1} = \lambda_{N+2} = 0$)

$$\Big(\frac{\partial R^m}{\partial y^m}\Big)^\top \lambda_m = \frac{\partial \mathcal{J}}{\partial y^m} - \Big(\frac{\partial R^{m+1}}{\partial y^m}\Big)^\top \lambda_{m+1} - \Big(\frac{\partial R^{m+2}}{\partial y^m}\Big)^\top \lambda_{m+2}, \tag{5.2}$$

and the total derivative

$$\frac{\mathrm{d}\mathcal{J}}{\mathrm{d}\theta} = \frac{\partial \mathcal{J}}{\partial \theta} - \sum_{n=1}^{N} \lambda_n^\top \frac{\partial R^n}{\partial \theta} - \Big(\lambda_1^\top \frac{\partial R^1}{\partial y^0} + \lambda_2^\top \frac{\partial R^2}{\partial y^0}\Big) \frac{\mathrm{d}y^0}{\mathrm{d}\theta} - \lambda_1^\top \frac{\partial R^1}{\partial y^{-1}} \frac{\mathrm{d}y^{-1}}{\mathrm{d}\theta}. \tag{5.3}$$

One forward and one backward sweep give all components, so the cost does not depend on $\dim \theta$ [9, 10]. Because we differentiate the discrete scheme ("discretize-then-optimize"), the gradient is the exact derivative of the computed $\mathcal{J}$, which is what gradient-based optimizers need and what a Taylor test can check to round-off [11]. For Dirichlet rows $\partial R_i^n / \partial y^{n-1} = 0$, which we implement with the interior mask $\mathsf{D}_\mathcal{I} = \mathsf{D}\,\mathrm{diag}(\mathbf{1}_\mathcal{I})$.

Taylor test. For a random direction d, $r_0(h) = |\mathcal{J}(\theta + hd) - \mathcal{J}(\theta)| = O(h)$ while $r_1(h) = |\mathcal{J}(\theta + hd) - \mathcal{J}(\theta) - h\nabla\mathcal{J} \cdot d| = O(h^2)$ *if and only if* the gradient is correct; any error in $\nabla\mathcal{J}$ degrades r_1 to $O(h)$.

5.2 Linear parabolic adjoint: conductivity, perfusion and source fields, Robin coefficient

The model $u_t - \nabla \cdot (\kappa \nabla u) + ku = Q$ is parameterized by

$$\kappa(\boldsymbol{x}) = \sum_j c_j \varphi_j(\boldsymbol{x}), \quad k(\boldsymbol{x}) = \sum_j p_j \psi_j(\boldsymbol{x}), \quad Q(\boldsymbol{x}, t) = Q_0(\boldsymbol{x}, t) + g(t) \sum_j s_j \chi_j(\boldsymbol{x}), \quad h \text{ (Robin)}. \tag{5.4}$$

Backward Euler with step k gives

$$R^n = (\mathsf{D} + k\mathsf{K}(\theta))u^n - \mathsf{D}u^{n-1} - k\big(\mathsf{D}Q^n(\theta) + b^n(\theta)\big), \qquad \mathsf{K} = \mathsf{A}(\kappa) + \mathsf{D}\,\mathrm{diag}(k(\boldsymbol{x}_i)) + \mathsf{B}(\kappa, h), \tag{5.5}$$

and (5.2) becomes a single *linear, constant* backward recurrence (one LU factorization of the transpose is reused):

$$(\mathsf{D} + k\mathsf{K})_\mathcal{B}^\top \lambda_m = \mathsf{P}^\top \mathsf{W}^m r^m + \mathsf{D}_\mathcal{I} \lambda_{m+1}, \qquad r^m = \mathsf{P}u^m - d^m, \tag{5.6}$$

where the subscript $\mathcal{B}$ denotes Dirichlet rows replaced by identity rows. With $\hat{\lambda}_n = \mathbf{1}_\mathcal{I} \odot \lambda_n$, the gradient components follow from (2.1):

$$\sigma_e := -k \sum_n \hat{\lambda}_n^\top \mathsf{G}_e u^n \quad \text{(edge sensitivity)}, \qquad \frac{\partial \mathcal{J}}{\partial c_j} = \sum_e \varphi_j(\boldsymbol{x}_e)\, \sigma_e + \sum_{f \subset \partial\Omega} \varphi_j(\boldsymbol{x}_f)\, \omega_f \frac{\partial \epsilon_f}{\partial \kappa_f}, \tag{5.7}$$

$$\frac{\partial \mathcal{J}}{\partial p_j} = -k \sum_n \sum_i \hat{\lambda}_{n,i} |\Omega_i| \psi_j(\boldsymbol{x}_i) u_i^n, \qquad \frac{\partial \mathcal{J}}{\partial s_j} = k \sum_n g(t_n) \sum_i \hat{\lambda}_{n,i} |\Omega_i| \chi_j(\boldsymbol{x}_i), \tag{5.8}$$

$$\frac{\partial \mathcal{J}}{\partial h} = \sum_f \omega_f \frac{\partial \epsilon_f}{\partial h}, \qquad \omega_f := -k \sum_n \hat{\lambda}_{n,i(f)} |f| \big(u_{i(f)}^n - u_\infty\big), \tag{5.9}$$

with, from (2.2),

$$\frac{\partial \epsilon_f}{\partial \kappa_f} = \frac{h^2 d_f}{(\kappa_f + h d_f)^2}, \qquad \frac{\partial \epsilon_f}{\partial h} = \frac{\kappa_f^2}{(\kappa_f + h d_f)^2}. \tag{5.10}$$

The edge sensitivities σ_e are accumulated once per time step with a single scatter over the sparse structure of A; any conductivity basis then costs only one dense product $\Phi^\top \sigma$. The gradient with respect to *every face conductivity* is therefore available at no extra cost.

5.3 Fractional (Caputo L1) adjoint

The L1 discretization of the Caputo derivative [6] is the scheme of `FractionalHeatSolver`.

The L1 scheme with $\varsigma = k^{-\beta}/\Gamma(2-\beta)$ and weights $b_\ell = (\ell+1)^{1-\beta} - \ell^{1-\beta}$ reads

$$R^m = \varsigma \mathsf{D} \sum_{j=0}^{m} c_j^{(m)} u^j + \mathsf{K} u^m - \mathsf{D} Q^m - b^m, \qquad c_j^{(m)} = b_{m-j}\,[j \geq 1] - b_{m-1-j}\,[j \leq m-1], \tag{5.11}$$

so $c_m^{(m)} = 1$ and every residual depends on the whole history. The adjoint inherits the memory, now *forward* in index:

$$(\varsigma \mathsf{D} + \mathsf{K})_{\mathcal{B}}^\top \lambda_m = \mathsf{P}^\top \mathsf{W}^m r^m - \varsigma \mathsf{D}_{\mathcal{I}} \sum_{\ell=m+1}^{N} c_m^{(\ell)} \lambda_\ell. \tag{5.12}$$

Using $\partial_\beta \varsigma = \varsigma\,(-\ln k + \psi(2-\beta))$ (ψ the digamma function) and $\partial_\beta b_\ell = -(\ell+1)^{1-\beta}\ln(\ell+1) + \ell^{1-\beta}\ln \ell$,

$$\frac{\partial \mathcal{J}}{\partial \beta} = -\sum_{m=1}^{N} \hat{\lambda}_m^\top \mathsf{D} \sum_{j=0}^{m} \big(\partial_\beta \varsigma\, c_j^{(m)} + \varsigma\, \partial_\beta c_j^{(m)}\big) u^j, \qquad \frac{\partial \mathcal{J}}{\partial \alpha} = -\sum_m \hat{\lambda}_m^\top \mathsf{A}_1 u^m, \tag{5.13}$$

where $\mathsf{A} = \alpha \mathsf{A}_1$. Forward and adjoint sweeps cost $O(N^2 M)$, like the forward L1 solver.

5.4 Nonlinear enthalpy (Stefan / Cattaneo–Stefan) adjoint

Here $y^n = H^n$, the residual is (3.9), and the observation is $\eta(H) = \mathcal{T}(H)$. From (3.10), $\partial R^{m+1}/\partial H^m = -p_{m+1}\mathsf{D}_{\mathcal{I}}$ and $\partial R^{m+2}/\partial H^m = -s_{m+2}\mathsf{D}_{\mathcal{I}}$, so (5.2) becomes

$$\mathsf{J}_m (H^m)^\top \lambda_m = \mathsf{S}(H^m)\, \mathsf{P}^\top \mathsf{W}^m r^m + p_{m+1} \mathsf{D}_{\mathcal{I}} \lambda_{m+1} + s_{m+2} \mathsf{D}_{\mathcal{I}} \lambda_{m+2}, \qquad r^m = \mathsf{P}\mathcal{T}(H^m) - d^m, \tag{5.14}$$

with the Newton Jacobian (3.13) at the converged state. The coefficients (γ_m, p_m, s_m) are those of the *accepted* steps, so the recurrence is valid on adaptive step sequences (the step-size selection is held fixed, the standard convention for adaptive discrete adjoints).

Latent heat. At fixed H, implicit differentiation of $H = cT + L\phi(T)$ gives

$$\frac{\partial \mathcal{T}}{\partial L}\Big|_H = -\frac{\phi(T)}{c_{\mathrm{app}}(T)}. \tag{5.15}$$

The contributions to $\mathrm{d}\mathcal{J}/\,\mathrm{d}L$ are: the explicit observation term $\sum_m (\mathsf{P}^\top \mathsf{W}^m r^m)^\top \partial_L \mathcal{T}(H^m)$; the residual term $-\sum_n \lambda_n^\top \mathsf{K}\, \partial_L \mathcal{T}(H^n)$ on interior rows and $+\sum_n \lambda_{n,\mathcal{B}}^\top \phi(g_{\mathcal{B}}^n)$ on Dirichlet rows (from $R_i = H_i - H(g_i; L)$); the initial-data term with $\mathrm{d}H^0/\,\mathrm{d}L = \phi(T^0)$; and, for $\tau > 0$, the start-up term with

$$\frac{\mathrm{d}H^{-1}}{\mathrm{d}L} = \phi(T^0) - k_0\, \phi'(T^0)\, \partial_t T^0 \Big(1 + \frac{k_0}{2\tau}\Big). \tag{5.16}$$

Diffusivity. $\partial R^n/\partial\alpha = \mathsf{A}_1\mathcal{T}(H^n)$ (interior rows) and $\mathrm{d}H^{-1}/\,\mathrm{d}\alpha = -\frac{k_0^2}{2\tau}\mathsf{D}^{-1}\mathsf{A}_1 T^0$.

Relaxation time. $\partial R^n/\partial\tau = \mathsf{D}D^2_{n-1}H$ (interior rows) and $\mathrm{d}H^{-1}/\,\mathrm{d}\tau = -\frac{k_0^2}{2\tau}a^0$.

With $L = 0$ this is the discrete adjoint of the Cattaneo heat equation; with $\tau = 0$ it is the adjoint of the classical enthalpy method.

6 Numerical results

All figures are produced by the two study scripts; numbers quoted below are read from their `summary.json` files.

6.1 Manufactured-solution verification of the enthalpy solver

We take the traveling front $T = A\tanh((x - x_0 - st)/w)$ with $A = 0.8$, $w = 0.18$, $s = 0.45$, $x_0 = -0.35$ on $[-1, 1] \times [-0.1, 0.1]$, the smooth law (3.5) with $L = 2$, $\delta = 0.25$, $\kappa = 0.08$, and the source

$$\tilde{Q} = \tau H_{tt} + H_t - \kappa T_{xx}, \qquad H_t = c_{\mathrm{app}}T_t, \quad H_{tt} = c_{\mathrm{app}}T_{tt} + L\phi''(T)T_t^2,$$

with exact Dirichlet data and exact $\partial_t T^0$ for the start-up level.

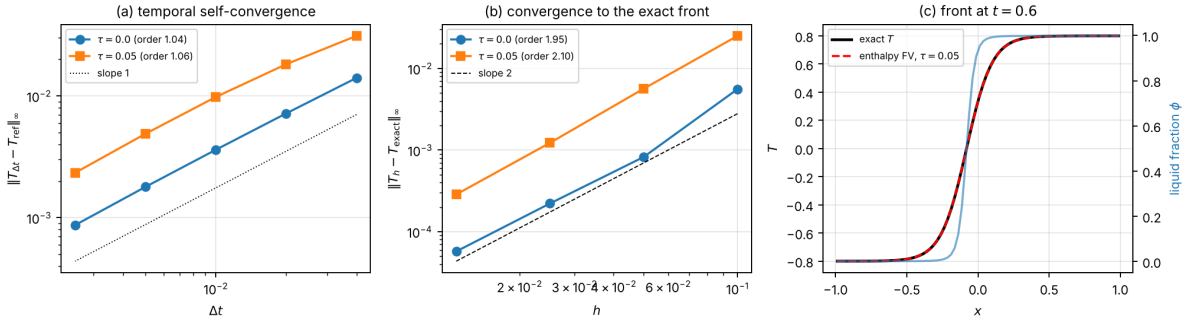


Figure 1: Manufactured traveling front. (a) Temporal self-convergence on an 80×8 mesh: order 1.04 ($\tau = 0$) and 1.06 ($\tau = 0.05$). (b) Convergence to the exact solution with $k \propto h^2$: order 1.95 and 2.10. (c) Computed and exact fronts with the liquid fraction.

Table 2: Observed orders (finest pair) for the enthalpy solver.

	temporal, $\max\lvert T_k - T_{\mathrm{ref}}\rvert$		spatial, $\max\lvert T_h - T\rvert$	
	finest error	order	finest error	order
Fourier ($\tau = 0$)	8.70×10^{-4}	1.04	5.74×10^{-5}	1.95
Cattaneo ($\tau = 0.05$)	2.34×10^{-3}	1.06	2.86×10^{-4}	2.10

6.2 Robustness for stiff mushy zones and energy conservation

A 20×20 unit square starts solid ($T = -0.5$) and is melted by a hot Dirichlet wall ($T = 1$ at $x < 0.05$), $\kappa = 0.05$, $L = 5$, $t_f = 0.5$. We sweep the smooth law's half-width $\delta \in \{0.2, \ldots, 0.002\}$ (capacity spikes 13.5 to 1251) and $k \in \{0.002, \ldots, 0.05\}$, and compare the enthalpy Newton solver with an apparent-capacity Picard solver given every advantage (Anderson depth 5, relaxation 0.7, up to 100 iterations).

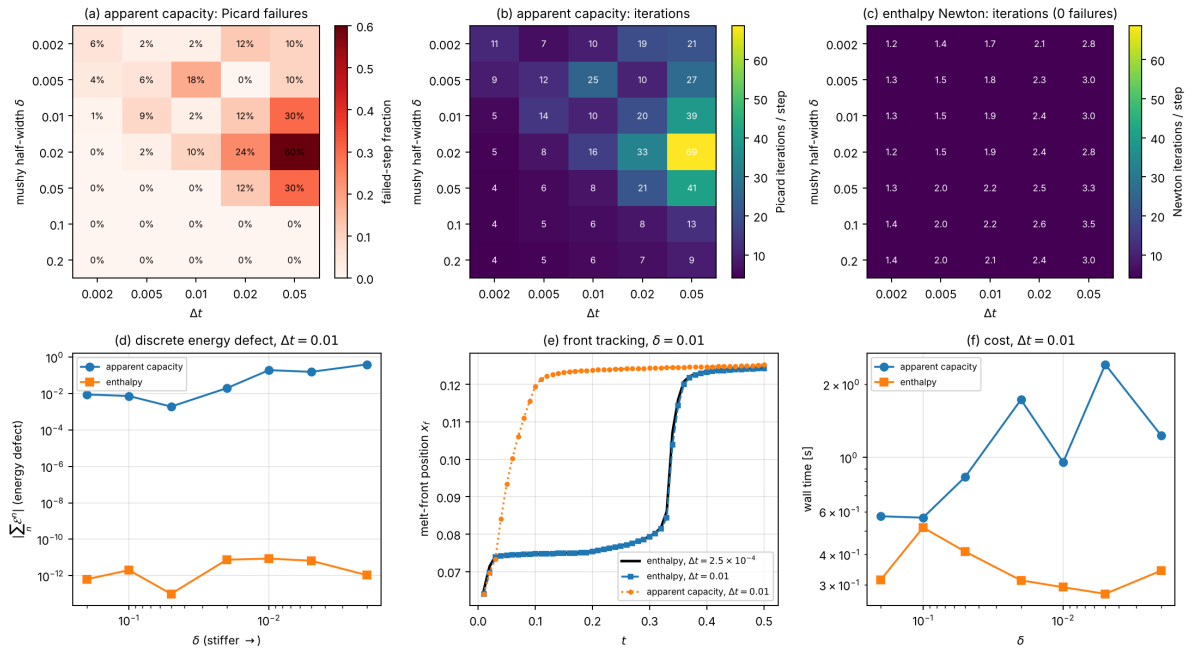


Figure 2: (a)–(c) Failed-step fraction and iterations per step over the $(\delta, \Delta t)$ grid. (d) Discrete energy defect (3.16) vs the enthalpy balance residual (3.15). (e) Melt-front position: the apparent-capacity front races ahead because its energy defect creates spurious latent heat. (f) Wall time.

Table 3: Stiffness sweep summary (35 runs per method).

	apparent capacity (Picard+Anderson)	enthalpy (Newton)
runs with non-converged steps	20 / 35 (up to 60% of steps)	0 / 35
iterations per step	4.1 – 68.7	1.2 – 3.5
$\lvert\sum_n$ energy defect$\rvert$	3.4×10^{-4} – 0.67	$\leq 1.5 \times 10^{-10}$
front error, $\delta = 0.01$, $\Delta t = 0.01$	4.9×10^{-2}	3.3×10^{-3}
total wall time (sum over sweep)	3.1×	1×

6.3 Adaptive time stepping on a multi-scale pulse problem

A 40×8 strip $[0, 1] \times [0, 0.2]$ at $T = -0.3$, ramp law with $T_s = -0.01$, $T_\ell = 0.01$, $L = 4$, receives an ultrafast flux pulse $q_{\text{in}} = 25$ for $t < 0.02$ on its left wall and then cools until $t = 2$ (time scales differ by two orders of magnitude). Errors are measured as the maximum over 61 output times of $\|T - T_{\text{ref}}\|_\infty$ against a uniform run with $k = 2 \times 10^{-5}$; work is the total number of Newton iterations including rejected steps.

At comparable work (≈400 Newton iterations) the adaptive error is 120× smaller; the adaptive run at rtol $= 3 \times 10^{-4}$ is 25× more accurate than the finest uniform run with 5× less work. We also observed (and the unit tests document) that on a *smooth, single-scale* diffusive problem per-step error control gives no advantage at equal step count: early errors are damped by diffusion, so spending steps on the initial transient does not pay off at the final time. Adaptivity is valuable precisely for the multi-scale forcing of laser processing.

6.4 Adjoint verification and cost

The unit tests additionally check (i) the enthalpy adjoint against central differences to relative accuracy 10^{-5} for $\tau \in \{0, 0.03\}$, Dirichlet and flux boundaries and a non-zero initial rate $\partial_t T^0$, (ii) that the field-adjoint forward solver reproduces `ReactionDiffusionHeatSolver` and the L1

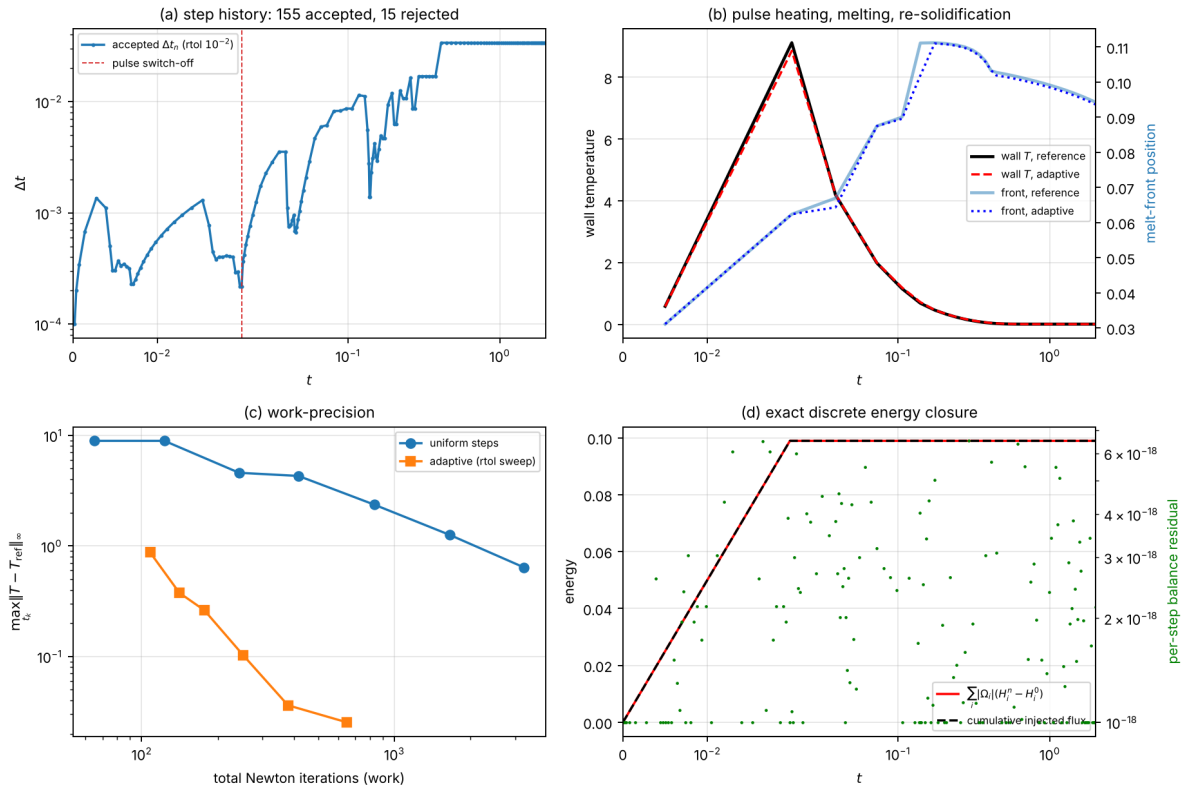


Figure 3: (a) Accepted step sizes (rtol $= 10^{-2}$): small during the pulse and at its switch-off, growing by two orders of magnitude during cooling (capped at the output spacing). (b) Wall temperature and melt front vs the reference. (c) Work–precision. (d) Total enthalpy gain equals the cumulative injected energy; per-step balance residuals are $O(10^{-18})$.

adjoint's forward solver reproduces `FractionalHeatSolver` to 10^{-12}, and (iii) that the gradient vanishes at the true parameters for noise-free data.

6.5 Conductivity-field inversion

The truth is $\kappa = 0.02 + 0.03\, G_{(0.32,0.66),0.12} + 0.02\, G_{(0.7,0.3),0.1}$ (Gaussian inclusions). It is *not* in the span of the inversion basis (a constant plus 49 Gaussians of width 0.08 on a 7×7 grid), so this is not an "inverse crime". Four heater experiments are observed by 49 sensors at 4 times with 1% Gaussian noise. We minimize the scaled misfit plus a smoothness penalty $\frac{\beta}{2}\,\|\mathrm{D}c\|^2 / c_{\mathrm{ref}}^2$ (D the grid difference operator on the bump coefficients, $\beta = 2 \times 10^{-4}$) with L-BFGS-B [12] (see [13] for the regularization background), starting from a uniform $\kappa = 0.03$.

6.6 Nonlinear parameter recovery

Five thermocouples in a 24×12 slab record 30 temperature samples while a boundary flux pulse melts it (enthalpy Stefan, smooth law, $\delta = 0.05$). From 1% noise and the initial guess $(L, \alpha) = (0.8, 0.09)$, L-BFGS-B with the enthalpy adjoint recovers (2.0023, 0.04998) against the truth (2, 0.05) in 13 iterations. The objective profiles in τ (Cattaneo–Stefan) and β (Caputo L1) show the adjoint derivatives as exact tangents.

Table 4: Work–precision data (fig. 3c).

	uniform			adaptive		
steps	Newton its	error	rtol	accepted / rejected	Newton its	error
200	243	4.58	10^{-1}	87 / 6	108	0.873
400	417	4.30	10^{-2}	155 / 15	177	0.262
800	831	2.36	3×10^{-3}	227 / 20	252	0.103
1600	1659	1.25	10^{-3}	351 / 23	379	0.0361
3200	3254	0.638	3×10^{-4}	611 / 31	646	0.0254

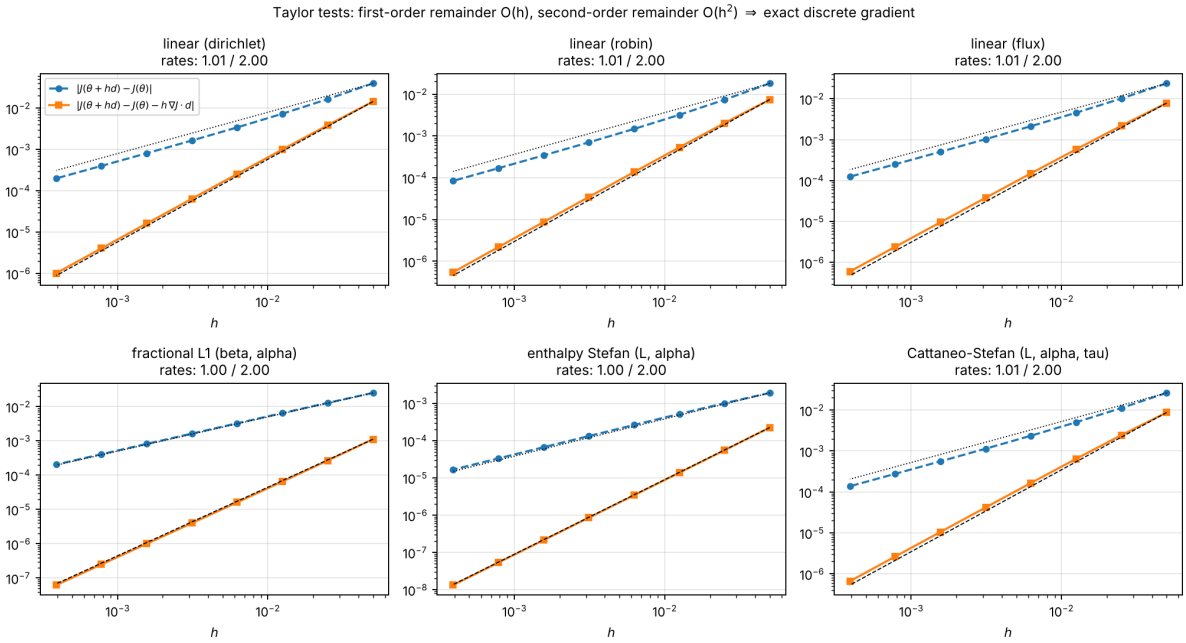


Figure 4: Taylor tests for all six adjoints. Dashed: $r_0 = O(h)$; solid: $r_1 = O(h^2)$. The finest-pair rates of r_1 are 2.00 for all six adjoints.

7 Limitations and next steps

- **Temporal order.** Backward Euler and the three-level Cattaneo scheme are first order in time (remark 3.2). A Newmark-averaged or BDF2 variant with an embedded error estimate would give second order; the adjoint framework (5.2) already handles three-level residuals.
- **Controller.** The elementary controller is adequate here; a PI controller [8] would smooth the step sequence after rejections. For $\tau > 0$ the estimator is a heuristic indicator.
- **Adjoint scope.** The enthalpy adjoint supports scalar α and Dirichlet/flux boundaries; extending it to conductivity fields reuses (5.7) with T in place of u. Robin and radiative boundaries in the nonlinear adjoint, and temperature-dependent conductivity, are not yet covered. The step sequence is not differentiated, so gradients on adaptive grids are exact for the realized grid only (standard practice).
- **Ramp law.** With the piecewise-linear law, $\mathcal{T}$ has kinks; Newton is semismooth there and line search keeps it robust, but gradients are only piecewise smooth. The smooth law is recommended for inversion.
- **Non-orthogonal meshes.** The M-matrix argument behind the Newton proposition requires a

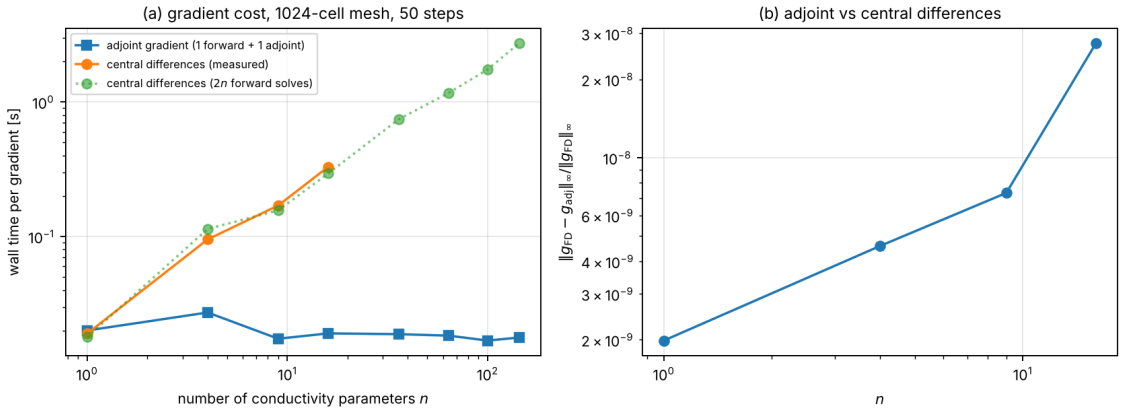


Figure 5: (a) Wall time per gradient on a 1024-cell mesh with 50 steps: the adjoint costs $\approx 0.02\,\mathrm{s}$ for 1 to 144 conductivity parameters; central differences grow linearly. (b) Relative difference to central differences is 10^{-9}–10^{-8} (finite-difference truncation).

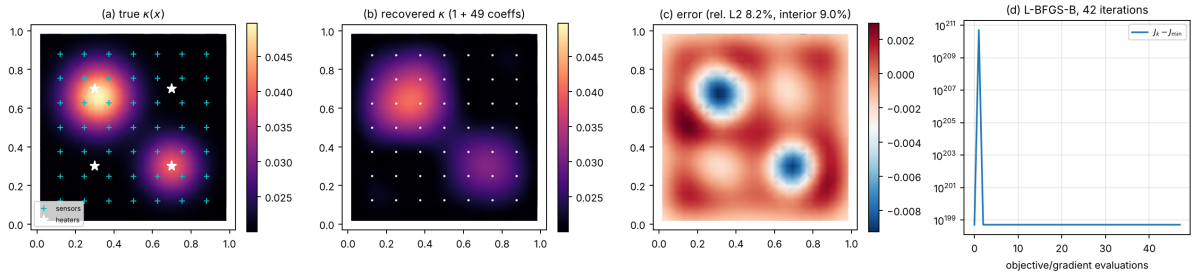


Figure 6: (a) True conductivity, sensors and heaters; (b) recovered field; (c) error; (d) optimizer convergence. The relative L^2 error falls from 34.9% (initial) to 8.2% after 42 L-BFGS-B iterations (48 gradient evaluations, a few seconds).

two-point (monotone) flux; with non-orthogonal corrections the Jacobian remains L-uniformly bounded but its M-matrix property is not guaranteed (see Direction B for monotone variants).

8 Reproducibility

```
python -m pip install -r requirements-dev.txt
export MPLCONFIGDIR=/tmp/matplotlib
python -m pytest tests/test_enthalpy.py tests/test_adjoint.py -q
python direction_c_enthalpy_studies.py   # ~5 min
python direction_d_adjoint_studies.py    # ~6 min
pdflatex direction_CD_enthalpy_adjoint_report.tex   # run twice
```

References

[1] V. R. Voller and C. Prakash, A fixed grid numerical modelling methodology for convection-diffusion mushy region phase-change problems, *Int. J. Heat Mass Transfer* 30 (1987) 1709–1719.

[2] V. R. Voller and C. R. Swaminathan, General source-based method for solidification phase change, *Numer. Heat Transfer B* 19 (1991) 175–189.

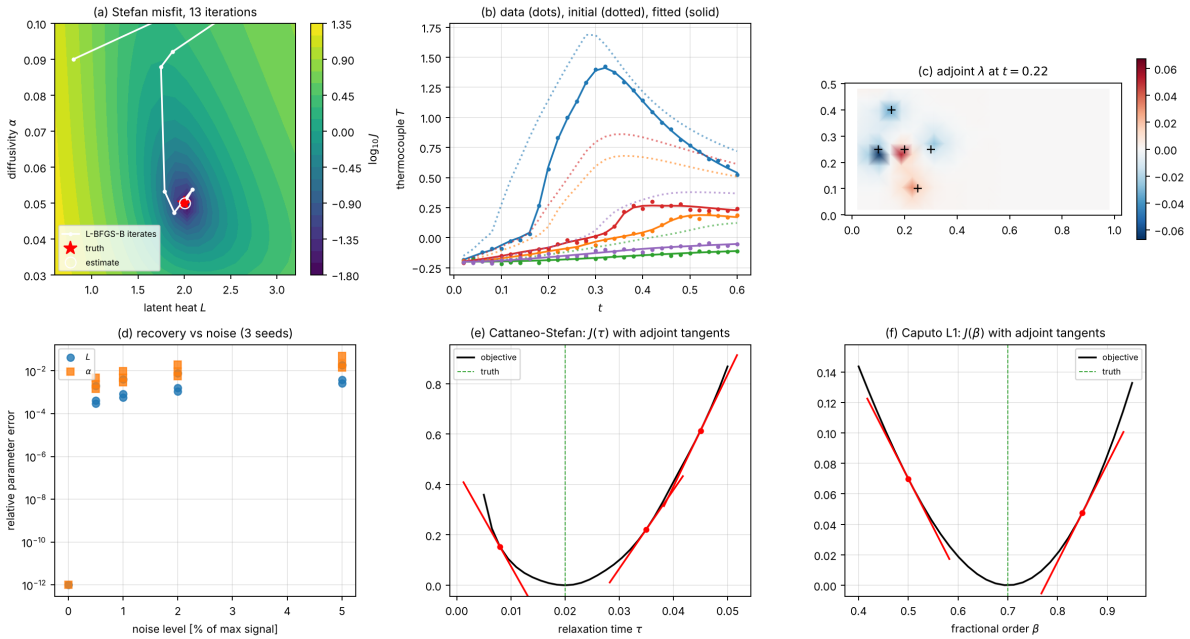


Figure 7: (a) Misfit landscape in (L, α) with accepted L-BFGS-B iterates. (b) Thermocouple data, initial prediction and fit. (c) Adjoint field λ, localized at the sensors and propagated backwards in time. (d) Relative parameter error vs noise level (3 noise seeds); noise-free recovery is exact to optimizer tolerance and errors grow roughly linearly with the noise. (e), (f) Objective profiles with adjoint tangent lines.

[3] C. M. Elliott, Error analysis of the enthalpy method for the Stefan problem, *IMA J. Numer. Anal.* 7 (1987) 61–71.

[4] C. Cattaneo, Sur une forme de l'équation de la chaleur éliminant le paradoxe d'une propagation instantanée, *C. R. Acad. Sci.* 247 (1958) 431–433.

[5] D. Y. Tzou, *Macro- to Microscale Heat Transfer: The Lagging Behavior*, 2nd ed., Wiley, 2015.

[6] Y. Lin and C. Xu, Finite difference/spectral approximations for the time-fractional diffusion equation, *J. Comput. Phys.* 225 (2007) 1533–1552.

[7] E. Hairer, S. P. Nørsett and G. Wanner, *Solving Ordinary Differential Equations I*, 2nd ed., Springer, 1993.

[8] G. Söderlind, Automatic control and adaptive time-stepping, *Numer. Algorithms* 31 (2002) 281–310.

[9] M. B. Giles and N. A. Pierce, An introduction to the adjoint approach to design, *Flow Turbul. Combust.* 65 (2000) 393–415.

[10] M. D. Gunzburger, *Perspectives in Flow Control and Optimization*, SIAM, 2003.

[11] P. E. Farrell, D. A. Ham, S. W. Funke and M. E. Rognes, Automated derivation of the adjoint of high-level transient finite element programs, *SIAM J. Sci. Comput.* 35 (2013) C369–C393.

[12] R. H. Byrd, P. Lu, J. Nocedal and C. Zhu, A limited memory algorithm for bound constrained optimization, *SIAM J. Sci. Comput.* 16 (1995) 1190–1208.

[13] C. R. Vogel, *Computational Methods for Inverse Problems*, SIAM, 2002.